CS5487 Lecture Notes (2024A)
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Tutorial 2

P-S 2.6: MLE for m.v. Gaussian

$$\log p(x_i) = -\frac{d}{2}\log 2\pi - \frac{1}{2}\log|\Sigma| - \frac{1}{2}\|x_i-\mu\|^2_{\Sigma}$$

For $D=\{x_1, \dots, x_n\}$

LL: $$\log p(D) = \sum_{i=1}^n \log p(x_i) = -\frac{nd}{2}\log 2\pi \underbrace{- \frac{n}{2}\log|\Sigma| \underbrace{- \frac{1}{2}\sum_{i=1}^n \|x_i-\mu\|^2_{\Sigma}}}$$

(a) MLE of μ:

$$\log p(D) \propto -\frac{1}{2}\sum_{i=1}^n \|x_i-\mu\|^2_{\Sigma} = -\frac{1}{2}\sum_{i=1}^n (x_i-\mu)^T\Sigma^{-1}(x_i-\mu)$$

$$= -\frac{1}{2}\sum_{i=1}^n \Big[\underbrace{x_i^T\Sigma^{-1}x_i}_{\times} - \underbrace{\mu^T\Sigma^{-1}x_i} - \underbrace{x_i^T\Sigma^{-1}\mu} + \mu^T\Sigma^{-1}\mu\Big]$$

"$(\mu^T\Sigma^{-1}x_i)^T = x_i^T\Sigma^{-1}\mu$" ← scalar: $a^T = a$

Take derivative:

$$\frac{\partial}{\partial\mu}\log p(D) = -\frac{1}{2}\sum_{i=1}^n \frac{\partial}{\partial\mu}\Big[-2x_i^T\Sigma^{-1}\mu + \mu^T\Sigma^{-1}\mu\Big]$$

Rules: $$\frac{\partial}{\partial x} a^Tx = \frac{\partial}{\partial x}\sum_j a_jx_j = \begin{bmatrix}\frac{\partial}{\partial x_1} \\ \vdots \\ \frac{\partial}{\partial x_d}\end{bmatrix}\sum_j a_jx_j = \begin{bmatrix}a_1 \\ \vdots \\ a_d\end{bmatrix} = a$$

univariate: $\frac{d}{dx}ax = a$; $\frac{d}{dx}ax^2 = 2ax$

$$\frac{\partial}{\partial x} x^TAx = Ax + A^Tx = 2Ax \quad \text{if } A \text{ symmetric.}$$

$$\frac{\partial}{\partial\mu}\log p(D) = \cancel{-\frac{1}{2}}\sum_{i=1}^n\left(-\cancel{2}\underbrace{\Sigma^{-1}x_i} + \cancel{2}\underbrace{\Sigma^{-1}\mu}\right) = 0$$

$$= -\sum_{i=1}^n \Sigma^{-1}x_i + N\Sigma^{-1}\mu = 0$$

$$\Rightarrow \Sigma^{-1}\mu = \frac{1}{N}\Sigma^{-1}\sum_{i=1}^n x_i \ \overset{\text{left-multiply}}{\Rightarrow}\ \text{w/ } \Sigma \Rightarrow \boxed{\mu = \frac{1}{N}\sum_{i=1}^n x_i}$$

(b) MLE for $\hat{\Sigma}$:

$$\log p(D) \propto -\frac{n}{2}\log|\Sigma| - \frac{1}{2}\sum_{i=1}^n \|x_i-\mu\|^2_{\Sigma}$$

$$= -\frac{n}{2}\log|\Sigma| - \frac{1}{2}\sum_{i=1}^n \underbrace{(x_i-\mu)^T}_{a}\underbrace{\Sigma^{-1}(x_i-\mu)}_{b}$$

trace trick: $\underset{\in\mathbb{R}}{c} = a^Tb = tr(a^Tb) = tr(ba^T)$

$$= -\frac{n}{2}\log|\Sigma| - \frac{1}{2}\sum_{i=1}^n tr\left(\Sigma^{-1}(x_i-\mu)(x_i-\mu)^T\right)$$

$$= -\frac{n}{2}\log|\Sigma| - \frac{1}{2}tr\left(\Sigma^{-1}\sum_{i=1}^n (x_i-\mu)(x_i-\mu)^T\right)$$

derivatives of matrix X

$$\frac{\partial}{\partial X}\log|X| = X^{-T} = X^{-1} \quad \text{if } X \text{ symmetric}$$

$$\frac{\partial}{\partial X}tr(X^{-1}A) = -(X^{-T}A^TX^{-T})$$

$$= -X^{-1}AX^{-1} \quad \text{if } A, X \text{ symmetric}$$

univariate: $\frac{\partial}{\partial x}\log x = \frac{1}{x} = x^{-1}$; $\frac{\partial}{\partial x}ax^{-1} = -ax^{-2}$

$$\Rightarrow \frac{\partial}{\partial\Sigma}\log p(D) = -\frac{n}{\cancel{2}}\Sigma^{-1} - \frac{1}{\cancel{2}}\left(-\Sigma^{-1}\Big[\sum_{i=1}^n (x_i-\mu)(x_i-\mu)^T\Big]\Sigma^{-1}\right) = 0$$

$$\underbrace{\Sigma\Sigma^{-1}\Sigma}_{\Sigma} \qquad \underbrace{\Sigma\Sigma^{-1}}_{I} \qquad \underbrace{\Sigma^{-1}\Sigma}_{I} \qquad \Sigma 0 \Sigma$$

left & right multiply w/ Σ

$$= -n\Sigma + \sum_{i=1}^n (x_i-\mu)(x_i-\mu)^T = 0$$

$$\Rightarrow \boxed{\hat{\Sigma} = \frac{1}{n}\sum_{i=1}^n (x_i-\mu)(x_i-\mu)^T}$$

PS 2.8: LS regression

input $x \in \mathbb{R}$

f.v.: $\phi(x) \in \mathbb{R}^D$

function: $f(x,\theta) = \phi(x)^T\theta \quad , \theta \in \mathbb{R}^D$

noisy version: $y = f(x,\theta) + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$

dataset: $D = \{(x_i, y_i)\}_{i=1}^n \Rightarrow$ goal: estimate param θ.

a) Least squares formulation:

$$LS = \sum_{i=1}^n \underbrace{(y_i - f(x_i,\theta))}_{\text{error}}{}^2 = \sum_{i=1}^n (y_i - \phi(x_i)^T\theta)^2$$

$$= \left\| \underbrace{\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}}_{y} - \underbrace{\begin{bmatrix} \phi(x_1)^T \\ \vdots \\ \phi(x_n)^T \end{bmatrix}}_{\Phi^T} \theta \right\|^2 = \| y - \Phi^T\theta \|^2$$

$\Phi = [\phi(x_1) \dots \phi(x_n)]$

minimize LS:

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \|y - \Phi^T\theta\|^2 = \underset{\theta}{\operatorname{argmin}} (y - \Phi^T\theta)^T(y - \Phi^T\theta)$$

$$= \underset{\theta}{\operatorname{argmin}}\ \cancel{y^Ty} - 2y^T\Phi^T\theta + \theta^T\Phi\Phi^T\theta$$

$a^Tb = b^Ta$

$a^Tb + b^Ta = 2a^Tb$

$$\frac{\partial}{\partial\theta} = -\cancel{2}\Phi y + \cancel{2}\Phi\Phi^T\theta = 0$$

$$\Rightarrow \Phi\Phi^T\theta = \Phi y$$

multiply by $(\Phi\Phi^T)^{-1}$ (assume invertible)

$$\Rightarrow \boxed{\hat{\theta} = (\Phi\Phi^T)^{-1}\Phi y}$$

b)

$$p(y_i | x_i, \theta) = N(y_i | f(x_i,\theta), \sigma^2)$$

LL: $$\log p(D) = \sum_{i=1}^n \log p(y_i | x_i; \theta)$$

$$= \sum_{i=1}^n \underbrace{-\tfrac{1}{2}\log 2\pi}_{\times} \underbrace{-\tfrac{1}{2}\log\sigma^2}_{\times} - \tfrac{1}{2}(y_i - f(x_i,\theta))^2$$

MLE:

$$\theta^* = \underset{\theta}{\operatorname{argmax}} \sum_{i=1}^n -\tfrac{1}{2}(y_i - f(x_i,\theta))^2$$

$$= \underset{\theta}{\operatorname{argmin}} \sum_{i=1}^n (y_i - f(x_i,\theta))^2 \Rightarrow \text{LS formulation}$$

$$\vdots$$
$$= (\Phi\Phi^T)^{-1}\Phi y$$

Notes: Explicit assumptions:

i) Gaussian Noise

ii) iid samples (iid noise)

iii) Zero mean & fixed variance

MLE describes other LS formulations

i) weighted LS (PS2-8) → different σ^2 for each x_i

ii) regularized LS (Lecture 3) → prior on θ.

iii) Lp-norm (PS2-9) → assume different noise models.